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GOGTE INSTITUTE OF TECHNOLOGY

UDYAMBAG, BELGAVI-590008

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Mini Project Report

TITLE

"Duster : Smart Automation for Modern Classrooms"

ELECTRICAL AND ELECTRONICS DEPARTMENT

Submitted by

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Department of Electrical and Electronics

CERTIFICATE

This is to certify that Mr Jay V Kangralkar, Mrs. Sahana I M, Mrs. Varsha Magdum, Mr. Yogiraj H of 5th semester, bearing USN 2GI23EE015, 2GI23EE038, 2GI23EE056, 2GI23EE061 has satisfactorily completed the Project . It can be considered as a bonafide work carried out in partial fulfilment for the academic requirement of 5th Semester B.E Electrical and Electronics Engineering prescribed by KLS Gogte Institute of Technology, Belagavi during the academic year 2025- 2026

The report has been approved as it satisfies the academic requirements in respect of Mini Project prescribed for the said Degree.

Signature of the Faculty Member Signature of the HOD

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ABSTRACT

The increasing demand for automation in educational and professional environments has created the need for efficient and hygienic alternatives to manual board-cleaning methods. Traditional dusters often lead to uneven cleaning, dust dispersion, classroom interruption, and reduced air quality. To address these limitations, this project presents the Smart Duster, an electromechanical system designed to automate the cleaning of whiteboards and blackboards. The prototype employs a dual-axis rail mechanism integrated with 500 RPM Johnson geared motors, limit switches for positional control, and a microcontroller-based control unit for precise movement and cleaning cycles. Powered by a 2200 mAh battery, the system ensures reliable operation, uniform cleaning, and reduced human effort. Market research and field surveys highlight the absence of commercially available automated board cleaners and the continued reliance on manual methods in most institutions. Experimental testing confirms smooth motion, accurate boundary detection, and effective cleaning performance on a 2 × 3 ft board. The Smart Duster demonstrates a cost-effective, energy-efficient, and user-friendly automation solution suitable for modern classrooms, with potential for further enhancement and commercialization.

1. INTRODUCTION

The increasing integration of automation into academic and professional environments has motivated the development of systems that enhance operational efficiency and hygiene. One such requirement arises in the frequent cleaning of whiteboards and blackboards used in classrooms, meeting rooms, and training centers. Manual board cleaning, though simple, often disrupts workflow, distributes dust particles, and results in inconsistent cleaning quality. These limitations highlight the need for an automated, reliable, and user-friendly solution.

The Smart Duster is an electromechanical automation system designed to perform systematic cleaning of writing boards without human intervention. The device employs a motor-driven mechanical assembly mounted on a guided track mechanism to achieve smooth traversal across the board surface. A microcontroller-based control unit drives the motion by regulating motor speed, stroke length, and direction while processing signals from limit switches or sensors for positional feedback. This integration of mechanical design, embedded control, and sensor-based automation ensures uniform cleaning, improved hygiene, and uninterrupted teaching or presentation activities.

The proposed system aims to provide an efficient and low-maintenance alternative to traditional dusters while demonstrating practical implementation of automation concepts such as motion control, microcontroller programming, and electromechanical system design. Its adaptability to various board dimensions and potential for further enhancements make it suitable for modern smart-classroom environments.

1.1 Objectives

The primary objectives of the Smart Duster project are as follows:

1. **To design and develop an automated board-cleaning mechanism** capable of performing consistent and uniform erasing across the entire board surface.
2. **To minimize human effort, physical strain, and dust exposure** by replacing manual board-cleaning operations with an automated solution.
3. **To implement an efficient mechanical design** incorporating a stable frame structure, smooth sliding mechanism, and appropriate duster mounting to ensure effective contact with the board.
4. **To build a low-cost, energy-efficient, and user-friendly system** that can be easily installed, operated, and maintained in classrooms, offices, and training centers.
5. **To evaluate system performance** through testing, calibration, and analysis of cleaning uniformity, mechanical stability, and control accuracy.

6. **To integrate a microcontroller-based control system** for precise regulation of motor motion, direction reversal, and cleaning cycles using sensor feedback.

1.2 Problem Statement

Conventional board-cleaning methods rely entirely on manual operation, which presents several challenges in academic and professional settings. Manual cleaning introduces frequent interruptions during lectures or presentations, thereby reducing teaching effectiveness and time utilization. Additionally, traditional dusters, especially those used on chalkboards, generate airborne dust particles that may cause respiratory discomfort, allergies, and reduced indoor air quality. Whiteboard markers often leave uneven stains when manually erased, affecting visibility and requiring repeated efforts.

The absence of a standardized cleaning technique further results in inconsistent cleaning patterns and inefficient coverage of the board surface. Moreover, for large boards, manual cleaning can be time-consuming and physically inconvenient. These issues collectively emphasize the need for an automated system that can deliver uniform cleaning quality, minimize human intervention, improve hygiene, and operate reliably across different board sizes.

Thus, the problem addressed by this project is the development of a **fully automated, microcontroller-based board-cleaning mechanism** that eliminates the drawbacks of manual erasing while enhancing operational efficiency, cleanliness, and user convenience.

2. LITERATURE REVIEW

2.1 Market Research

A thorough market study was conducted to identify the availability, commercial adoption, and technological evolution of automated board-cleaning systems. The research revealed that while several academic projects and prototypes exist in the domain of automated whiteboard/blackboard cleaners, **there are no widely available commercial products for institutional purchase**. Most existing implementations remain in the prototype stage and are not mass-manufactured or standardized for market deployment.

Several research papers explored advanced concepts such as **vacuum-assisted dust removal**, **rotary brushes**, **robotic arm–based cleaning mechanisms**, and **rail-based wiping systems**, indicating growing academic interest but limited industrial translation. However, these designs either lack cost-effectiveness or require complex mechanical assemblies, making

them unsuitable for practical classroom environments. This market gap highlights a clear opportunity for developing a low-cost, functional, and easily deployable Smart Duster.

2.2 Survey

As part of the survey, visits were conducted to a few educational institutions including schools and colleges. Observations indicated that traditional manual cleaning methods are predominantly used, regardless of the board type (whiteboard or blackboard). In less-developed or resource-constrained institutions, the quality of dusters was poor—often made from cloth scraps or improvised materials—resulting in inefficient cleaning and accumulation of residual ink or chalk dust.

Faculty members reported frequent interruptions during teaching, uneven cleaning quality, and dust-related discomfort. The absence of automated solutions across all surveyed institutions validates the necessity for a simple, effective, and low-cost automated board-cleaning device. The survey findings strongly support the relevance and utility of the Smart Duster system.

3. METHODOLOGY

The Smart Duster was developed using a structured electromechanical design methodology, integrating mechanical movement, sensor feedback, and microcontroller-based control.

A set of **horizontal railings** was mounted on the **top and bottom edges** of a 2 × 3 ft board to support the translation of the cleaning mechanism. Additionally, **two vertical rods** were installed, enabling **dual-axis movement**—vertical (up–down) motion and horizontal (left–right) motion—of the duster carriage. The mechanical structure allows the duster to also move **slightly front and back**, ensuring proper contact pressure with the board surface.

The movement of the rods is controlled using **500 RPM Johnson geared motors**, selected for their high torque and reliability. **Limit switches** were integrated at boundary positions to detect extremities of motion and prevent mechanical overrun. These switches serve as input signals to the microcontroller, enabling controlled direction reversal and precise cleaning cycles.

A **2200 mAh rechargeable battery** powers the entire system, supplying adequate current to the motors and microcontroller while ensuring portability. The control logic was implemented through embedded programming, where motor actuation, limit-switch handling, and cleaning sequence execution were coded to achieve smooth bi-directional movement along both axes.

This methodology ensures a mechanically stable, electronically controlled, and energy-efficient cleaning mechanism suitable for classroom use.

4. WORKING/OPERATION

When the system is powered ON, the microcontroller initializes the motor driver and monitors all limit-switch inputs. The cleaning cycle begins with the horizontal movement of the duster carriage along the top and bottom railings. As the motor rotates, the duster moves across the board until it reaches a limit switch at the end of the horizontal axis. Upon activation, the microcontroller reverses the motor direction.

Simultaneously, the vertical rods enable up–down movement, allowing the duster to sweep the entire surface area of the 2 × 3 ft board. Each time a vertical limit switch is triggered, the controller adjusts the duster's height, ensuring complete and uniform coverage. The slight front–back movement ensures consistent pressure, improving wiping efficiency.

The cleaning continues until the defined cycle is completed or until the user manually stops the system. The operation integrates **motor control**, **feedback loops**, and **mechanical translation**, providing a fully automated and controlled cleaning process.

5. SELECTION OF MATERIALS

5.1 Electrical & Mechanical Calculations (Torque, Power, Current, Energy)

1. Assumptions

- Board dimension: **2 ft × 3 ft** → width used for horizontal sweep = **3 ft = 0.9144 m**.
- Carriage + duster mass (estimated) = **m = 1.50 kg**
- Coefficient of sliding friction (or equivalent resistance force factor) between carriage and guide = **μ = 0.10**
- Gravity acceleration = **g = 9.81 m/s²**.
- Drum/pulley radius used to convert motor rotation to linear motion = **r = 0.020 m (20 mm)**.
- Motor no-load speed (given) = **500 rpm**.
- Selected gearbox ratio (to reduce speed and increase torque at output) = **GR = 10 : 1** (common small gearbox; you may use other ratios).

- Gearbox mechanical efficiency (typical small gearboxes) = **η_gear = 0.80 (80%)**.
- Motor electromechanical efficiency under load = **η_motor = 0.70 (70%)** (conservative).
- Battery nominal voltage assumed = **V = 12 V** (typical for Johnson DC motors).
- Battery capacity given = **2200 mAh = 2.2 Ah**.

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2. Force and required torque at the output drum

Frictional (resistive) force estimate:

Using a simple friction model (resistive force approximated as μ·m·g)

$$F = \mu \cdot m \cdot g$$

Plugging numbers:

$$F = 0.10 \times 1.50 \times 9.81 = 1.4715 \text{ N}$$

Torque at output drum (before safety margin)

Torque required to overcome this force at drum radius $r = 0.02$m:

$$T_{output,base} = F \times r = 1.4715 \times 0.02$$
$$= 0.02943 \text{ N} \qquad \text{\cdotpm}$$

Design / safety margin

Allow a safety margin (to cover acceleration, extra friction, misalignment). Use factor of **2**:

$$T_{output,required} = 2 \times T_{output,base} = 2 \times 0.02943$$
$$= 0.05886 \text{ N} \qquad \text{\cdotpm}$$

Interpretation: The cleaning system should be able to deliver ≈ **0.059 N·m** at the output drum to move the carriage reliably.

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3. Rotational speeds and linear speed

Motor angular speed (500 rpm → radians/s):

$$\omega_{motor} = 500 \times \frac{2\pi}{60} = 52.35987756 \text{ rad/s}$$

Output angular speed after gearbox (GR = 10):

$$\omega_{out} = \frac{\omega_{motor}}{10} = 5.235987756 \text{ rad/s}$$

Linear speed at drum:

$$v = \omega_{out} \times r = 5.235987756 \times 0.02 = 0.10472 \text{ m/s}$$

Time to cross the 3-ft (0.9144 m) width at that speed:

$$t_{cross} = \frac{0.9144}{0.10472} = 8.7319 \text{ s}$$

So ≈ **8.73 seconds** per horizontal sweep (one pass).

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4. Mechanical power required (output shaft)

Mechanical power at output (using required torque and output angular speed):

$$P_{mech} = T_{output,required} \times \omega_{out}$$
$$P_{mech} = 0.05886 \text{ N} \qquad \text{\cdotpm}$$
$$\times 5.235987756 \text{ rad/s} = 0.30819 \text{ W}$$

So ≈ **0.31 W mechanical** power is required at the gearbox output to overcome friction and accelerate the carriage under our assumptions.

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5. Electrical input power and current (approximate)

Account for motor efficiency ($\eta_{motor} = 0.70$):

$$P_{elec} = \frac{P_{mech}}{\eta_{motor}} = \frac{0.30819}{0.70} = 0.44027 \text{ W}$$

Assuming battery voltage $V = 12$ V:

$$I = \frac{P_{elec}}{V} = \frac{0.44027}{12} = 0.03669 \text{ A} \approx 37 \text{ mA}$$

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6. Battery runtime estimate

Using theoretical current above:

$$\text{Battery energy / runtime (ideal)} = \frac{2.2 \text{ Ah}}{0.03669 \text{ A}} = 59.96 \text{ hours}$$

This extremely long runtime is a consequence of the low computed steady current. For realistic runtime estimation, use a conservative practical operating current. For example, assume a typical operating current under load of **I_{practical} = 1.0 A** (common ballpark for small DC motors under moderate load and during starts):

$$\text{Runtime}_{1A} = \frac{2.2 \text{ Ah}}{1.0 \text{ A}} = 2.2 \text{ hours}$$

If motor draws 0.5 A on average:

$$\text{Runtime}_{0.5A} = \frac{2.2}{0.5} = 4.4 \text{ hours}$$

Recommendation: use motor datasheet stall/nominal current to compute realistic runtime. For safety and design, assume **0.5–1.0 A** average draw to set realistic runtime expectations (2.2–4.4 hours for a 2.2 Ah battery).

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7. Energy per sweep (useful for duty-cycle estimates)

Energy consumed electrically during one horizontal sweep (time ≈ 8.7319 s):

$$E_{elec_per_sweep} = P_{elec} \times t_{cross} = 0.44027 \text{ W} \times 8.7319 \text{ s} = 3.8444 \text{ J}$$

This is a small amount of energy per sweep (≈3.84 J) under the theoretical loading; again, real energy will be higher when accounting for dynamic acceleration and start/stop losses.

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9. Summary

- Estimated carriage mass: **1.50 kg** (assumption — measure and replace).
- Frictional force estimate: **1.4715 N**.
- Required output torque (with ×2 margin): **0.05886 N·m**.
- Selected gearbox: **10:1** → output angular speed **5.236 rad/s**, linear speed **0.1047 m/s**.
- Mechanical power required at output: **0.308 W**.
- Electrical input power (η_motor = 70%): **0.440 W**.
- Theoretical steady current at 12 V: **≈0.037 A (37 mA)** — treat as theoretical lower bound.
- Practical expected average current for small DC motor under load: **0.5–1.0 A** → battery runtime **~2.2–4.4 hrs** (for 2.2 Ah battery).
- Energy per single sweep (electrical): **≈3.84 J** (theoretical).

6. RESULTS AND DISCUSSIONS

The developed Smart Duster successfully demonstrated reliable and consistent board-cleaning performance on a 2 × 3 ft surface. The rail-based design ensured smooth horizontal and vertical motion, while limit switches provided accurate positional control. The 500 RPM Johnson motor offered sufficient torque to move the rods and duster assembly without stalling.

Testing indicated uniform cleaning with minimal streaks. The autonomous movement reduced interruptions during classroom activity and eliminated dust exposure associated with manual dusters. The 2200 mAh battery provided adequate operational duration for multiple cleaning cycles without frequent recharging.

However, certain observations emerged during evaluation. The alignment of railings is critical for smooth motion; minor misalignments may introduce friction. The cleaning efficiency depends on the pressure applied by the carriage, which may require fine-tuning for different board surfaces. Despite these challenges, the prototype proves the feasibility of a cost-effective automated board-cleaning solution.

7. ADVANTAGES AND DISADVANTAGES

Advantages:

- Fully automated and reduces manual effort
- Ensures uniform and consistent cleaning
- Minimizes dust exposure and improves hygiene
- Portable and battery-powered
- Cost-effective compared to advanced robotic cleaners
- Sensor-based operation enhances reliability and safety

Disadvantages:

- Mechanical Wear and Alignment Issues
- Dependence on Electrical Components
- Limited Compatibility and Cleaning Pressure

8. APPLICATIONS

- Schools and colleges
- Smart classroom installations
- Training centers and coaching institutes
- Office meeting rooms
- Conference and seminar halls
- Public presentation spaces

9. CONCLUSION

The Smart Duster project successfully demonstrates a practical and affordable solution to modernize board-cleaning practices. Through the integration of mechanical railings, dual-axis motion mechanisms, limit-switch feedback, high-torque motors, and microcontroller-based automation, the system delivers efficient and uniform cleaning on a 2 × 3 ft board.

Market research and surveys confirm the absence of commercial alternatives and highlight the need for such automation in educational institutions. The prototype effectively reduces

manual effort, improves hygiene, and enhances the overall teaching experience. The project establishes a strong foundation for future enhancements such as wireless control, improved cleaning mechanisms, and commercial scalability.

10. REFERENCE

Websites & Online Resources

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